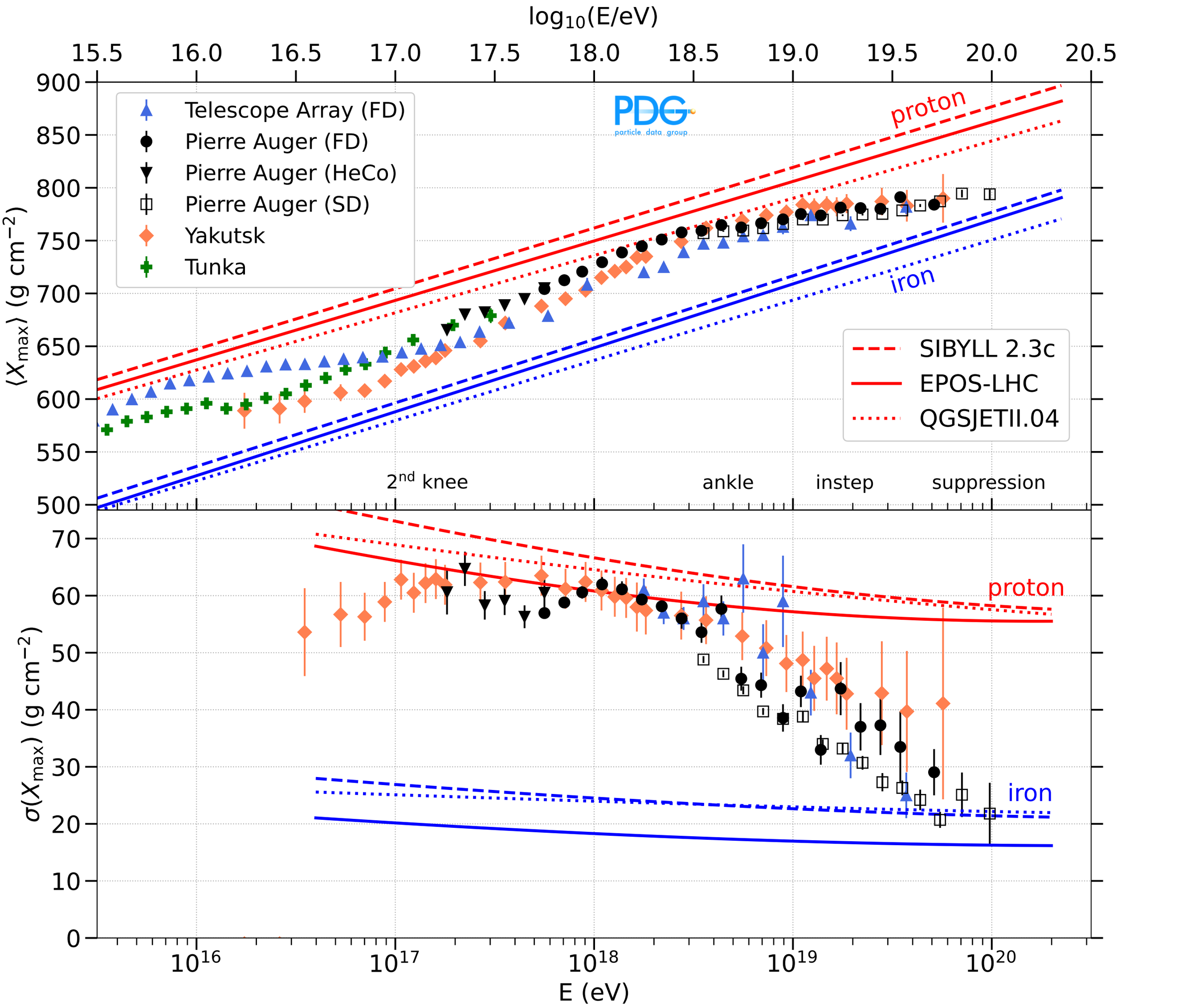
log10(E/eV)
15.5
16.0
16.5
17.0
17.5
18.0
18.5
19.0
19.5
20.0
20.5
⟨Xmax⟩ (g cm−2)
900
850
800
750
700
650
600
550
500
Telescope Array (FD)
Pierre Auger (FD)
Pierre Auger (HeCo)
Pierre Auger (SD)
Yakutsk
Tunka
PDG
particle data group
proton
iron
SIBYLL 2.3c
EPOS-LHC
QGSJETII.04
2nd knee
ankle
instep
suppression
σ(Xmax) (g cm−2)
70
60
50
40
30
20
10
0
proton
iron
10^16
10^17
10^18
10^19
10^20
E (eV)